## 1. CHEMICAL REACTIONS AND EQUATIONS SOLUTIONS

1. State any three observations which help to determine whether a chemical reaction has taken place. Write an activity which may justify at least two of the listed observations.

Answer:

Any three observations indicating a chemical reaction:

- Change in state (solid/liquid/gas) or formation of a gas.
- Change in colour.
- Change in temperature (heat evolved/absorbed) or light produced.

Activity (shows gas evolution and temperature change):

1. Take a few zinc granules in a test tube.
2. Add dilute hydrochloric acid (dil. HCl) to it.
3. Bring a burning splinter near the mouth of the test tube.

Observations:

- Bubbles of gas are evolved (hydrogen gas).
- The test tube becomes warm (reaction is exothermic).
- The gas burns with a 'pop' sound confirming hydrogen.

Chemical equation: $Zn + 2HCl \rightarrow ZnCl_2 + H_2\uparrow$

2. Mention three ways by which a usual chemical equation can be made more informative.

Answer:

A chemical equation can be made more informative by adding:

- Physical states of reactants and products: (s), (l), (g), (aq).
- Conditions required for the reaction such as temperature/heat, pressure, catalyst, light, electricity, etc.
- Energy change: whether heat is evolved (exothermic) or absorbed (endothermic) (often written as +Heat/−Heat).

3. Consider the following reaction: $Pb(NO_3)_2(s) \xrightarrow{Heat} PbO(s) + NO_2(g) + O_2(g)$

(a) Name the gases produced. (b) Balance the equation. (c) Name the type of reaction.

Answer:

(a) Gases produced: nitrogen dioxide ($NO_2$) and oxygen ($O_2$).

(b) Balanced equation:

$2Pb(NO_3)_2(s) \xrightarrow{Heat} 2PbO(s) + 4NO_2(g) + O_2(g)$

(c) Type of reaction: thermal decomposition reaction (decomposition on heating).

4. (a) When hydrogen gas burns in presence of oxygen, water is formed and when water is electrolysed then hydrogen and oxygen gases are produced.

State the kind of reaction that takes place (i) in the first case (ii) in the second case.

(b) In electrolysis of water, hydrogen and oxygen are produced at cathode and anode respectively. Mention the ratio of volumes of hydrogen and oxygen.

Answer:

(a)

- (i) Hydrogen burning in oxygen to form water is a combination reaction (also an exothermic oxidation/combustion reaction).

  $2H_2 + O_2 \rightarrow 2H_2O$ + heat

- (ii) Electrolysis of water is a decomposition reaction (electrolytic decomposition).

  $2H_2O \xrightarrow{electricity} 2H_2 + O_2$

(b) Volume ratio of hydrogen : oxygen = 2 : 1 (since $H_2$ is produced in double the volume of $O_2$).

5. Food when left for a long time tastes/smells bad. What is this condition called? Mention any four ways by which we can prevent this condition.

Answer:

The condition is called rancidity (oxidation of fats and oils in food).

Methods to prevent rancidity (any four):

- Keep food in airtight containers to prevent contact with oxygen.
- Store food in a refrigerator/at low temperature.
- Add antioxidants (e.g., BHA, BHT) to food items.
- Use nitrogen flushing for packing chips and other oily foods.